Yanyu Wang

Earth Commons Postdoc Fellow
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EDUCATION

University of Maryland Center for Environmental Science 2021-2026
Ph.D. in Marine, Estuarine, and Environmental Sciences (MEES)

- Dissertation title: "Toward circular and sustainable nitrogen management in the US and globally"
- Advisor: Dr. Xin Zhang

Emory University- Atlanta, GA 2019-2021
M.S. in Environmental Sciences

- Thesis title: "Agricultural Greenhouse Gas Fluxes Under Different Cover Crop Systems"
- Advisor: Dr. Eri Saikawa

Colorado State University - Fort Collins, CO 2017-2019
B.S. in Ecosystem Science and Sustainability

Anhui Agriculture University - Hefei, China 2015-2017
B.S. in Environmental Science

HONORS AND GRANTS

- Ann G. Wylie Dissertation Fellowship ($15,000) – University of Maryland Spring 2025
- Debbie Morrin-Nordlund Memorial Award ($1,600) – University of Maryland Summer 2024
- Dean's Fellowship ($10,000) – University of Maryland Summer 2024
- Lester Research Grants ($5,000) – Emory University Summer 2020
- The Water Scholars Program Scholarship ($50,000) – Coca-Cola August 2017

RESEARCH EXPERIENCE

Postdoc Fellow – Earth Commons, Georgetown University Sep 2026- Present

- Investigate how crop–livestock systems influence nitrogen losses, greenhouse gas emissions, and air pollution across the United States.
- Develop spatial optimization strategies for manure recycling and crop–livestock recoupling to advance a circular nutrient economy.
- Advisor: Dr. Megan Lickley

Summer intern – Jornada experimental range, NM June to July 2025

- Contributed to manure recycling research by assisting with manure application on farms and conducting greenhouse experiments
- Contributed to research on manure excretion and recycling estimates as part of the *Nitrogen 2.0* project, funded by Spark Climate Solutions
- Advisor: Dr. Sheri Spiegal

Graduate Research Assistant – UMCES-Appalachian Lab July 2021 to Jan 2026

- Lead the research work for a NSF funded project: "INFEWS: U.S.-China: Managing agricultural nitrogen to achieve sustainable Food-Energy-Water Nexus in China and the U.S."
- Advisor: Dr. Xin Zhang

OTHER PROFESSIONAL EXPERIENCE

Early career/young scientist convener – American Geophysical Union December 2024

- Promoted the scientific session through social media and coordinated outreach to increase participation.
- Facilitated networking opportunities for early-career researchers and invited speakers during the session.

Stakeholder workshop organizer – Institute of Marine and Environmental Technology November 2022

- Coordinated stakeholder recruitment for one of the first parallel U.S.–China workshops on sustainable nitrogen management, engaging participants from academia, government, industry, and non-governmental organizations.
- Co-developed the workshop agenda and facilitated discussions that informed stakeholder-engaged research and international collaboration on sustainable nitrogen management.

Teaching Assistant – Emory University August 2020 to May 2021

- Facilitated breakout discussions; prepared answer keys, programming assignments, exams, and course projects; graded coursework; and held weekly office hours for Climate Change and Society (ENVS 326/526; Dr. Eri Saikawa) and Fundamental Concepts in Soil Science (ENVS 285; Dr. Debjani Sihi).

ESS Club International Coordinator – Colorado State University Sep 2019 - May 2019

- Developed and coordinated activities that increased engagement of international students in the Ecosystem Science and Sustainability (ESS) Club.

Undergraduate researcher, USDA-ARS Fort Collins Jan to Dec 2018

- Evaluated the performance of the COMET-Farm greenhouse gas prediction model by comparing model outputs with field measurements under the supervision of Dr. Steve Del Grosso and Dr. Catherine Stewart.

PUBLICATIONS

- **Wang, Y.**, B. Gu, X. Zhang. Shifting U.S.–China trade from feed to food reduces global agricultural nitrogen loss and greenhouse gas emissions. Nitrogen Cycling 2, nc-0026-0011 (2026). https://doi.org/10.48130/nc-0026-0011
- **Wang, Y.**, X. Zhang, S. Spiegal, E. A. Davidson. A framework for estimating manure nitrogen balance and recycling potential for current and future conditions in the USA. Nature Food 7, 260–271 (2026). https://doi.org/10.1038/s43016-026-01312-5
- Zhang, X., R. Sabo, L. Rosa, H. Niazi, P. Kyle, J. Byun, **Y. Wang**, X. Yan, B. Gu, E. Davidson. Nitrogen management during decarbonization. Nature Reviews Earth & Environment 5, 717–731 (2024). https://doi.org/10.1038/s43017-024-00586-2
- Zhang, X., **Y. Wang**, L. Schulte-Uebbing, W. de Vries, T. Zou, E. A. Davidson. Sustainable Nitrogen Management Index: Definition, Global Assessment and Potential Improvements. Frontiers of Agricultural Science and Engineering 9(3), 356–365 (2022). https://doi.org/10.15302/J-FASE-2022458
- **Wang, Y.**, E. Saikawa, A. Avramov, N. S. Hill. Agricultural Greenhouse Gas Fluxes Under Different Cover Crop Systems. Frontiers in Climate 3, 742320 (2022). https://doi.org/10.3389/fclim.2021.742320

CONFERENCE PRESENTATIONS

Wang, Y. *Integrating systems thinking into nitrogen management: Insights from the U.S. Nitrogen Budget (2012–2022)*. AGU Fall Meeting. New Orleans, LA. 2025.

Wang, Y. *Assessing manure nitrogen balance and recycling potential with a structured framework for current and future conditions*. AGU Fall Meeting. New Orleans, LA. 2025.
Wang, Y. *Nitrogen budget in the contiguous United States and its decadal shifts from 2012 to 2022*. American Geophysical Union (AGU) Fall Meeting. Washington, DC. December 2024.
Wang, Y. *Shifting trade from feed to food reduces agricultural nitrogen loss and greenhouse gas emissions in the U.S. and China*. Invited Seminar, University of Illinois Urbana–Champaign. Champaign, IL. July 2024.
Wang, Y., X. Zhang, E. A. Davidson. *Reintegrating livestock and crop production to achieve a sustainable Energy–Water–Land (EWL) nexus*. Oral Presentation, American Geophysical Union (AGU) Fall Meeting. San Francisco, CA. December 2023.
Wang, Y. *Importing feed or food? Environmental impacts of shifting agricultural trade between the United States and China*. XXI International Nitrogen Workshop. Madrid, Spain. October 2022.
Wang, Y. *Agricultural greenhouse gas fluxes under different cover crop systems*. Oral Presentation, American Geophysical Union (AGU) Fall Meeting. December 2020.
Wang, Y. *Trace gas fluxes under different cover crop systems*. 7th Annual Southeastern Biogeochemistry Symposium. March 2020.
Wang, Y. *Evaluation of the COMET-Farm decision support system and the DayCent ecosystem model using field observations*. 24th Celebrate Undergraduate Research and Creativity Symposium. Colorado State University. April 2018.